

Is the extremal slope new?

On the manuscript's $\partial M_{\text{ext}}/\partial \alpha|_J$, compared with Kleihaus, Kunz & Radu, arXiv:2303.12471 (Ref. [2]).

Not ready to submit as it is.

Ref. [2] already contains $M_{\text{ext}}(\alpha)$ and its slope, over about six times our coupling range. Our numbers agree with it. The paper needs reframing before it goes to a journal.

1. M_{ext} is the observable; its slope adds little

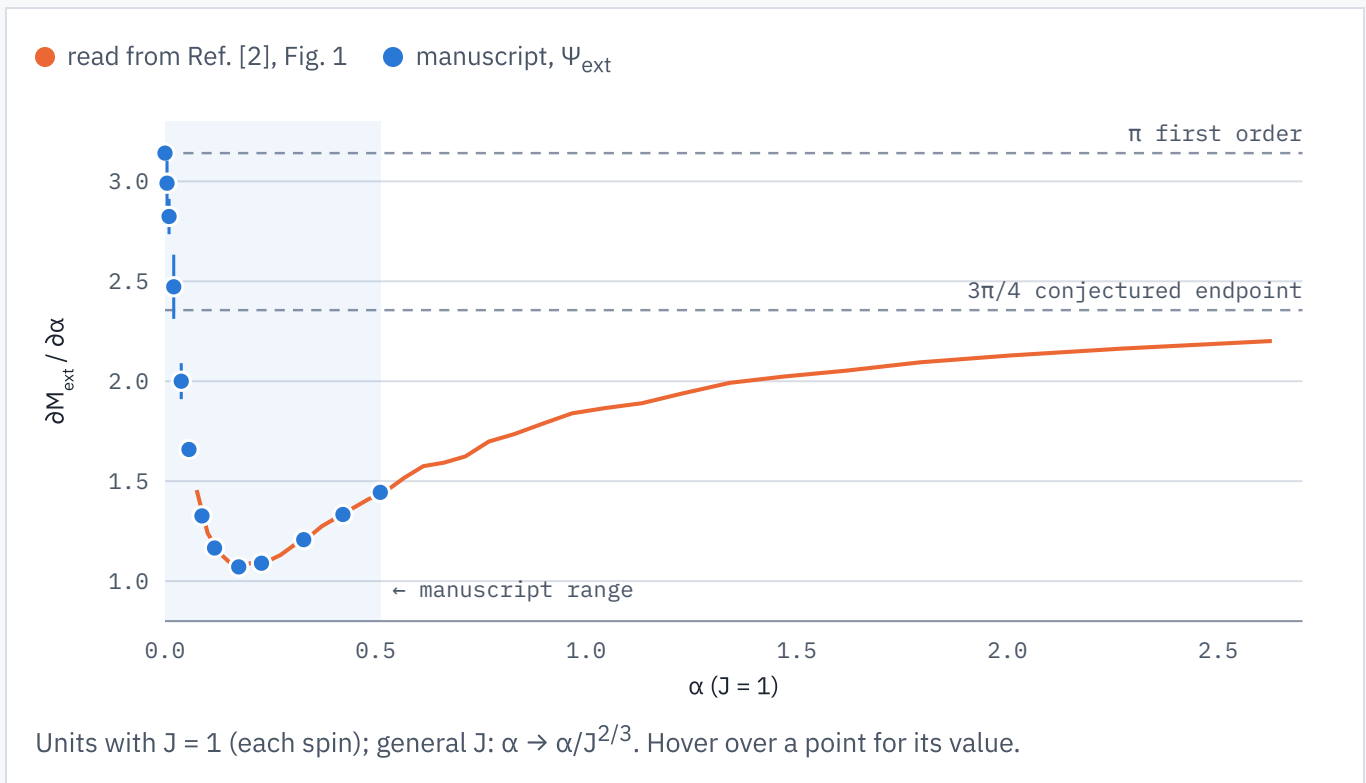
$M_{\text{ext}}(J, \alpha)$ is the physically relevant quantity. It is the extremality bound itself: it says which (M, J) a black hole can have at a given coupling. Its slope, compared with M_{ext} , carries little new information:

- The physics in the slope is its sign (the bound tightens) and its shape. Both are visible in $M_{\text{ext}}(\alpha)$.
- The slope is the natural object only at $\alpha \rightarrow 0$, where first-order perturbation theory gives π . At finite α , the comparison with perturbation theory is cleaner on M_{ext} itself. At $y = 0.51$ the first-order formula predicts a shift 2.3 times the measured one, a mass 32% too high.
- Its links to the entropy (Goon–Penco) and to Ψ in the extended first law are consistency identities. The manuscript itself declines to draw weak-gravity conclusions for uncharged, rotating holes. Ψ would be physical if α were set by a field such as a modulus, but the manuscript does not argue this.
- Its practical value is technical: Ψ gives the slope pointwise, without differencing. That makes it a good tool and check for M_{ext} , but not a result in itself.

2. Both can be read from Ref. [2]

How. Ref. [2] prints no M_{ext} , but its Fig. 1 plots the extremal horizon area against α/M (main panel) and against $J/M^{3/2}$ (inset). Matching the area between the two curves gives $J/M^{3/2}$ as a function of α/M on the extremal branch, which is $M_{\text{ext}}(J, \alpha)$. I took the curves from the PDF's vector data, and the entropy panels confirm the result.

Result. It agrees with ours to 2×10^{-3} in M_{ext} and 0.015 in the slope, minimum included. Ref. [2] reaches $\alpha \approx 2.6$; we stop at 0.51.



$\alpha (J = 1)$	M_{ext} ours	M_{ext} Ref. [2]	$\partial M_{\text{ext}}/\partial \alpha$ ours	$\partial M_{\text{ext}}/\partial \alpha$ Ref. [2]
0.088	2.3735	2.3727	1.326	1.352
0.175	2.4740	2.4733	1.071	1.079
0.329	2.6466	2.6463	1.208	1.205
0.511	2.8885	2.8884	1.444	1.437

3. Why Ref. [2] never stated it

This is my inference. Ref. [2] is a short survey of three families of solutions. Its question was whether the Einstein bound $j \leq 1$ survives. That is weaker than knowing $M_{\text{ext}}(\alpha)$: it only says the bound never relaxes. Like most higher-dimensional phase diagrams, its plots are normalised by the mass, so $M_{\text{ext}}(J)$ never appears as a plotted quantity. Ref. [2] does not cite the first-order result of Ma, Li and Lü, so it had no perturbative value to compare against. These authors also rarely give data tables.

4. Is it worth sending to a journal?

Not in its current form. Its headline (M_{ext} grows, the slope dips and then rises) can be read from a figure in Ref. [2], and a referee who knows that paper would object. The abstract's claim that Ref. [2] gives "not $M_{\text{ext}}(J, \alpha)$ or its slope" does not survive this comparison.

Yes, after two changes.

- Reframe the paper as a precise, error-controlled confirmation of Ref. [2]. Lead with what Ref. [2] cannot give: Ψ on non-extremal solutions and its sign change, the implicit-response method, and the numerical check of Ma–Li–Lü. In that form it is a solid but modest paper for PRD or CQG.
- For a stronger result, take the branch past $x = 0.92$ and test Ref. [2]'s conjecture that it ends at $M = 3\pi\alpha/4$ with $J \rightarrow 0$. That is the open question Ref. [2] leaves. Earlier work in this repo already reaches $x \approx 0.81$.

Reproduce: `python experiments/kkr_fig1_extremal_mass.py`. Data in `results/kkr_fig1-extremal-mass.json`; full version in `report.pdf`. Sections 3 and 4 are judgement, not computation.